

Measuring Digital Adoption: Global Indexes

- Understanding how countries adopt digital technology
- Why digital adoption matters for economic growth
- Overview of global measurement frameworks

What is Digital Adoption?

Integration of digital tools into:

- Everyday life
- Business operations
- Government services

Not just technology availability → *effective usage*

Key areas:

- Internet usage
- Digital payments
- Online services
- Automation & cloud tools

Integration of digital tools into government services

- **Online shopping** (Daraz, Amazon, Flipkart)
- **Digital payments** (Khalti, eSewa, FonePay QR everywhere)
- **Navigation apps** (Google Maps for routes, traffic)
- **Telemedicine** (online doctor consultations)
- **E-learning** (YouTube classes, Coursera, Google Classroom)
- **Social media communication** (Messenger, WhatsApp, Instagram)

Integration of digital tools into business operations

- **Online tax filing** (IRIS portal in Nepal, GST portal in India)
- **Digital ID systems** (Nagarik App, Aadhaar)
- **Online passport application**
- **Digital land records** (LRO systems, online maps)
- **Online payments for utilities** (electricity, water, tax)
- **E-governance portals** for complaints, certificates, permits
- **Online education services** from public schools/universities

Integration of digital tools in daily life

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Key Areas

Internet usage

- % of people using mobile data daily
- Watching news, social media, search engines
- Students accessing online learning

Online services

- Booking tickets (airlines, buses)
- Ordering food online
- Government certificates online

Digital payments

- QR codes in shops (FonePay, Khalti QR)
- Mobile banking transfers
- Paying bills online

Automation & cloud tools

- Google Workspace for collaboration
- Automated email marketing
- Cloud servers instead of local servers
- AI chatbots for customer service

Why Measuring Digital Adoption Matters

- Shows how digitally advanced a country is
- Helps identify infrastructure gaps
- Helps governments plan digital policies
- Helps businesses see market opportunities
- Used by World Bank, UN, OECD, IMF, ITU
- **Example:**
Countries with high digital adoption perform better during crises (COVID remote work, online schooling).

World Bank's Digital Adoption Index (DAI)

- Global framework to measure adoption across **3 pillars**:
- People
- Business
- Government
- Used to compare digital readiness across countries
- Helps determine digital transformation progress

DAI Pillar 1: People

- **Measures:**

- Access to internet
- Smartphone usage
- Social media adoption
- Online learning & telemedicine
- Digital payments usage

- **Examples:**

- Nepal: TikTok penetration extremely high
- India: 300M UPI transactions per day
- South Korea: nearly universal high-speed internet

DAI Pillar 2: Businesses

- **Measures:**

- Use of cloud platforms
- Ecommerce participation
- Online advertising
- Digital accounting systems
- Automated supply chains

- **Examples:**

- Amazon uses AI-driven logistics
- Nepali SMEs rarely use cloud accounting
- Restaurants using Foodmandu/Pathao Food = digital adoption

DAI Pillar 3: Government

- **Measures:**

- Online public service delivery
- Digital ID systems
- Tax filing systems
- Open data portals
- E-governance platforms

- **Examples:**

- Estonia: 99% government services online
- India: Aadhaar + DigiLocker
- Nepal: Nagarik App + Online passport appointment

OECD Digital Government Index (DGI)

- **What it measures:**
 - How modern, efficient, and citizen-focused a digital government is
 - Goes beyond access → measures *quality* and *maturity*
- **Used by:**
 - OECD countries (Europe, USA, Japan, etc.)

DGI Dimension 1: User-Centricity

- Services designed for people, not for bureaucracy
- Simple UI, fast processing, mobile-friendly
- Example:
 - UAE: 90% services available on one app
 - Estonia: one-click tax filing
 - **Relatable:**
When a government website actually works smoothly → user-centric.

DGI Dimension 2: Digital By Design

- Government systems built digitally from the start
- Not “paper system → digital copy”
- Unified platforms for identity, payments, signatures
- **Examples:**
 - IndiaStack (Aadhaar, eKYC, UPI)
 - Singapore’s SingPass

DGI Dimension 3: Data-Driven Public Sector

- Government uses data to:

- Improve decisions
- Predict trends
- Provide smarter services

Examples:

- Singapore uses data to manage bus frequency
- India uses data for digital health missions
- Google Mobility data used in COVID response worldwide

DGI Dimension 4 & 5

4. Proactiveness

- Govt proactively delivers services
- Example:
 - Denmark automatically sends child benefits after birth registration

5. Government as a Platform

- Shared govt building blocks (ID, payments, cloud)
- Example:
 - UPI in India used by thousands of apps
 - Estonia's X-Road connects all govt systems

DGI Dimension 6: Open By Default

- Transparency, public access to government data
- Encourages innovation & accountability
- **Examples:**
 - UK's open data portal
 - USA govt APIs
 - Covid data dashboards shared openly

Nepal's Digital Adoption Status

- **People: Moderate–High**
- TikTok, Facebook, Instagram usage very high
- Digital payments growing (eSewa, Khalti, FonePay)
- But internet quality inconsistent
- **Business: Low–Moderate**
- Digital accounting rare
- Cloud adoption low
- Ecommerce rising but informal

Nepal's Digital Adoption Status contd.

- **Government: Early Stage**
- Some progress: Nagarik App, online passport
- But:
 - Systems not integrated
 - Manual paperwork still dominates
 - No digital-by-design approach

Summary & Discussion

- Digital adoption shapes competitiveness
- Global indexes help benchmark progress
- Nepal improving but gaps remain in:
 - Business digitalization
 - Government integration
 - Infrastructure & policy
 - **Discussion Questions:**
- Which pillar (people/business/government) is Nepal strongest in?
- Why does business digital adoption lag behind?
- What can government learn from Estonia or India?